PSTD 400 · SENIOR SEMINAR · OCTOBER 2026

Pennsylvania’s 7th, precinct by precinct

Our forecast of the 2026 House race: Bob Brooks (D) vs. Ryan Mackenzie (R)

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INTRODUCTIONS

Meet the team

PROJECT LEAD

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TODAY

Four questions, in order

1

The forecast

Who wins, by how much, and how sure we are

2

Main features

The data, the model, and the variables that matter

3

What is distinctive

What our approach adds over simpler forecasts

4

Against Polymarket

Where we disagree with the market, and why

Mackenzie holds the seat by about 2.5 points

Ryan Mackenzie (R)

51.3%

156,517 votes

Bob Brooks (D)

48.7%

148,738 votes



Margin: Mackenzie by 7,779 votes, about 2.5 points of the two-party vote

How much weight that number can bear

0.13 pts
District-level miss
predicting 2024 vote shares using only 2022 and earlier data

1.9 pts
Average miss per precinct
same 2024 backtest, across 335 comparable precincts

48.7–49.2%
Brooks share across our two models
the precinct model and a separate municipal check

What the backtest does not test: turnout, because it counted votes using actual 2024 turnout, or a midterm year.

What we cannot give yet: a win probability.

Carbon outweighs Lehigh, and Northampton tips it

County	Brooks	Mackenzie	Brooks share	Net margin
Carbon	9,200	17,553	34.4%	Mackenzie +8,353
Lehigh	72,292	67,529	51.7%	Brooks +4,763
Monroe (part)	1,777	3,672	32.6%	Mackenzie +1,895
Northampton	65,469	67,763	49.1%	Mackenzie +2,294
District	148,738	156,517	48.7%	Mackenzie +7,779

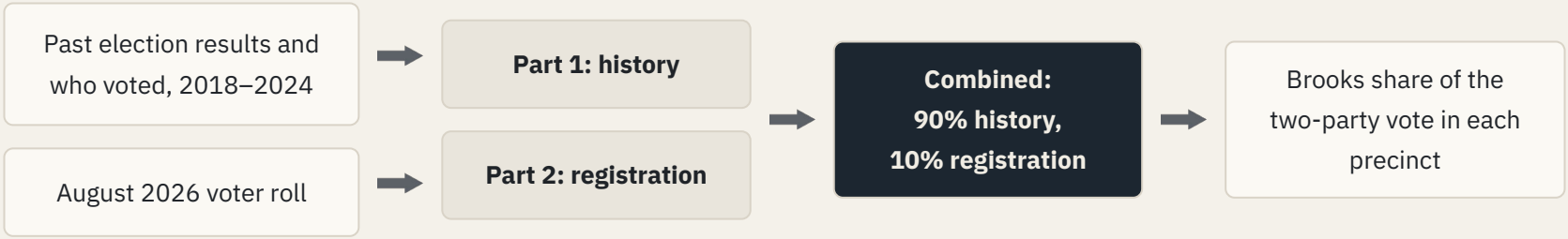
Brooks wins the median precinct (50.7%) but loses the district (48.7%). The district share is total Brooks votes over total votes, and the precincts Mackenzie carries cast more votes.

2

Main features

The data, the structure of the model, what we tried and dropped, and which variables carry the prediction.

One forecast model, in two parts



Vote counts are arithmetic, not the model: each precinct's share × its 2022 House votes, plus 2.76% for registration growth, summed to the district.

The municipal model is a separate check. It is not averaged into the forecast.

Why 2026 registration gets 10%

History 90%

Registration 10%: one-tenth of how far the forecast moves from precinct to precinct comes from the 2026 roll

Why not more

Past results alone explained about 97% of 2024. The roll counts everyone registered, including people who rarely vote.

Why not zero

It is the only data dated 2026, so it is how changes since 2024 enter the forecast.

What we have not done

Tested other weights. 10% is a judgment call; next we backtest 0%, 5%, 10% and 20%.

How past elections are weighted

Weighting means averaging. A 67/33 weight makes each input 0.67 × its 2022 value + 0.33 × its 2024 value.

LEARN

2018 → 2020

2018 inputs predict 2020 results

→

LEARN

2020 → 2022

2020 inputs predict 2022 results

→

BACKTEST

2020 + 2022 → 2024

80% 2022, 20% 2020: the most recent election counts most

→

FORECAST

2022 + 2024 → 2026

67% 2022, 33% 2024: the last midterm counts most

Both weightings were judgment calls, and the backtest used a different rule from the forecast. We will rerun it with one rule.

Two data sources carry the model

Certified precinct results, 2018–2024

President, Senate, Governor and House, for Carbon, Lehigh, Northampton and the Monroe precincts in the district.

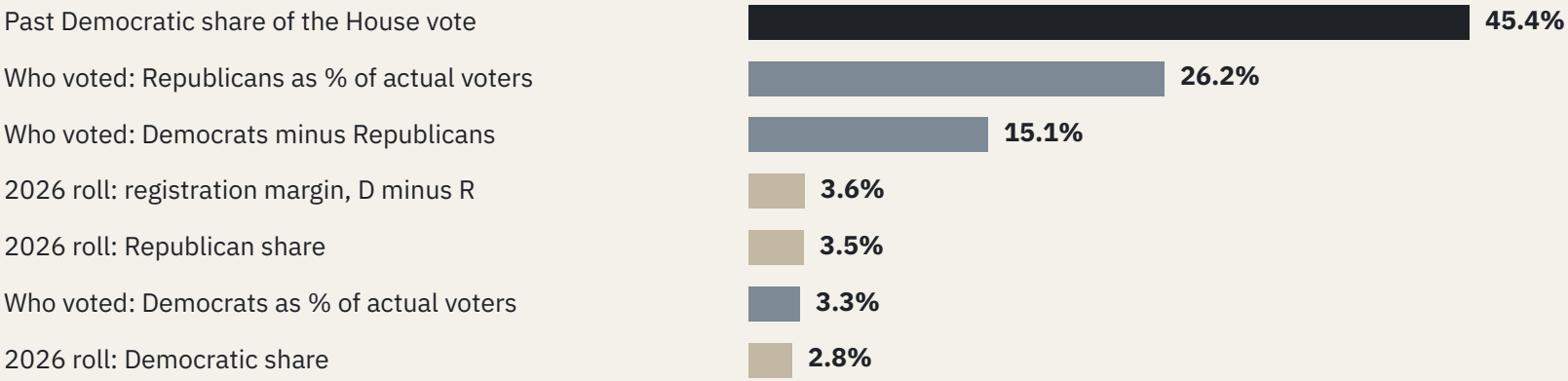
County voter files, as of Aug 31, 2026

Who is registered, with which party, age and sex, and whether each person voted in past elections, never for whom.

Fixes that changed the numbers

- Nazareth and Lower Nazareth were swapped; Carbon’s 2022 and 2024 columns were misaligned
- Six Northampton towns were redrawn for 2026 (27 precincts became 32); each is one row in training
- Precincts whose votes were reported inside a neighbor were checked against voter-file turnout

Past results and who voted do most of the work



Results: how the precinct voted

Who voted: party registration of people who cast ballots, from the voter file, never how they voted

2026 roll: everyone registered now

Every feature had to beat random noise

7 / 28

candidate features survived

1. Add 10 columns of pure random numbers to the data
2. Rebuild the forest 20 times
3. Keep a feature only if it beats every random column in at least 19 of 20 rebuilds and carries 5% of its forest
4. Drop the failures and repeat until every survivor passes

The weakest survivor was still 11 times larger than the best random column.

Earlier methods we tested, and set aside

Approach	What happened
Trend, beta coefficient, regression	First-round methods; beta-adjusted won the 2024 backtest
Gradient boosting, elastic net	Less accurate than the random forest
X and Reddit sentiment	No gain: 2022 vote alone explained about 97% of 2024
Campaign spending	Studied as a coefficient; not in this version
Separate turnout forecast	Dropped; the model forecasts share only
Municipal-level model	Kept as an independent cross-check

None of these feeds the forecast. The forecast comes from one model: the precinct random forest.

3

What is distinctive

What a precinct-level, voter-file model adds over a district-wide number or a repeat of 2024.

A forecast you can audit precinct by precinct

Geographic detail

370 precinct forecasts that add up exactly to the district total. Every vote in the headline can be traced to a place.

Turnout composition

The voter file shows which partisans actually turned out, not just how a place voted. Those inputs carry over 40% of the model.

Tested inputs

Every variable beat random noise, and each one’s contribution is measured in votes, not asserted.

The payoff is a map of relative strength: which precincts should run ahead of or behind the district. That is what a district-wide swing would be applied to.

Why we lead with the precinct model

2024 backtest and 2026 forecast	Precinct	Municipal
District-level miss	0.13 pts	0.31 pts
Average miss per town, 86 shared towns	1.45 pts	1.52 pts
Rows the model learns from	370 precincts	89 towns
2024 rows with a comparable result	335 of 377	89 of 89
2026 forecast, Brooks share	48.7%	49.2%

The edge is narrow. The precinct model is noisier row by row, but more accurate once its rows are added up.

4

Against Polymarket

The market and our model point in opposite directions. The interesting question is why.

The market says Brooks; our model says Mackenzie

Polymarket, Oct 6

86%

chance Brooks wins

about $34,000 traded; Kalshi at 85%

Franklin & Marshall poll, Sept 9–20

45 – 40

Brooks 45%, Mackenzie 40%

about 53% Brooks in two-party terms

Our model

48.7%

Brooks two-party share

Mackenzie favored by about 2.5 points

Sources: Polymarket and Kalshi, Oct 6, 2026; Franklin & Marshall poll via PoliticsPA, Sept 24, 2026

They are answering different questions

What our model sees

- How precincts voted relative to each other, 2020–2024
- Which partisans turned out
- August 2026 registration

What the market also prices

- The national environment and midterm penalty
- District polling
- Candidates, money, and who is on the ballot

Agree: the district’s baseline is close to even. **Disagree:** which way 2026 swings off that baseline.

Is this a trading edge? Not yet

Buying Mackenzie at 14 cents is a bet that 2026 looks like 2020–2024 in this district.

Our model assumes that. It has never tested it. The 2024 backtest was a presidential year, and we have no midterm backtest or probability yet.

Before the final forecast

1	**Redo the backtest**	One weighting rule for backtest and forecast, predicted turnout, and a 2022 midterm test
2	**Test the registration weight**	Backtest 0%, 5%, 10% and 20% and keep the best
3	**Add a swing layer**	Generic ballot, midterm history and polls
4	**Model turnout**	Use 2026 mail ballot requests and returns instead of flat 2.76% growth
5	**Produce a probability**	Uncertainty bands from backtest errors across several cycles

WHERE WE STAND, OCTOBER 2026

Mackenzie 51.3 · Brooks 48.7

A strong precinct baseline that still needs a swing. Questions?